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18.05.2026(E)

Your Roll No.....

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Sr. No. of Question Paper : 7471
Unique Paper Code : 2233520003
Name of the Paper : DSC-22: Faunal Conservation and Restoration
Name of the Course : **B.SC. (H) ZOOLOGY/B.SC (P) LIFE SCIENCE
THEORY - 2026**
Semester : VIII (NEP-UGCF)

Duration : 3 Hours

Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt any five questions in all.
3. Question No. 1 is compulsory.

1. (a) Define the following:

(1x5)

- (i) Metapopulation
- (ii) Biomagnification
- (iii) Rewilding
- (iv) Invasive species
- (v) DNA barcoding

(b) Differentiate between the following:

(2x5)

- (i) National Park and Biosphere Reserve
- (ii) Overexploitation and habitat destruction
- (iii) Bioaccumulation and Biomagnification
- (iv) Ecological restoration and conservation
- (v) Sentinel Species and Keystone Species

(c) Expand the following:

(1x5)

- (i) NTCA
- (ii) WWF
- (iii) EIA
- (iv) NBWL
- (v) UNEP

P.T.O.

(1x5)

(d) Fill in the blanks:

- (i) The _____ is the world's first national park, established in 1872.
- (ii) The phenomenon by which pollutants increase in concentration up the food chain is called _____.
- (iii) Species found only in a particular geographic area are called _____ species.
- (iv) The spread of non-native species into new ecosystems is studied under _____ ecology.
- (v) Wildlife forensics uses _____ protocols for species identification from feathers, bones, or hair

(1x5)

(e) State true or false:

- (i) Genetic diversity refers to the variety of ecosystems present in a region.
- (ii) The Western Ghats are recognized as one of India's biodiversity hotspots.
- (iii) Sentinel species act as environmental guardians by indicating ecosystem health.
- (iv) Ecological economics ignores ethical values while evaluating development projects.
- (v) The release of GMOs into the environment may pose ecological risks.

2. (a) What is genetic diversity? Give a detailed account of faunal biodiversity in the northeastern region of India. (7)

(b) Discuss the significance of faunal biodiversity hotspots in India, with special reference to the Himalayan region, Western Ghats, and North-Eastern region. How do these regions contribute to global biodiversity conservation? (8)

3. (a) Explain invasive species ecology with two case studies from India. (7)

(b) Describe how global climate change is one of the major threats to biodiversity with suitable examples. (8)

4. (a) Discuss the important biotic and abiotic factors for ecological restoration. (7)

(b) Explain ecological restoration with reference to any two major biotic communities. (8)

5. (a) What is wildlife forensics? Explain forensic protocols for species identification from different parts of reptiles, birds, and mammals. (7)

(b) Give two wildlife crime case studies and explain the role of forensics in solving them. (8)

6. (a) Explain how pollution ecology (air, water, soil, and radioactive) poses threats to biodiversity. Give suitable examples (7)

(c) Describe the concepts of GIS and remote sensing. Discuss their role in ecological monitoring with examples. (8)

7. Write short notes on any three of the following: (3x5)

- (i) Captive breeding and reintroduction programs
- (ii) Radioactive pollution and fauna
- (iii) Bioprospecting and biopiracy
- (iv) Environmental impact assessment